

Aqeel Ahmed, PhD

Data Scientist | Quantitative Researcher

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🇪🇺 Permanent Residence Permit (Germany)



Summary

PhD physicist with 10+ years of experience in high-dimensional quantitative modeling, statistical analysis, and large-scale computational simulations using Python/C++, backed by 30+ peer-reviewed publications (950+ citations). Applying this quantitative foundation to deliver Data Science and Machine Learning/AI solutions.

Key Skills

ML & Data Science	NumPy, Pandas, Scikit-learn, PyTorch
Programming/Tools	Python, SQL, C/C++, Git, Google Cloud, Docker
Domain Strengths	Predictive Modeling, Scientific Communication
Languages	English (Fluent), German (Intermediate B1/B2)

Experience

- **Max-Planck-Institut für Kernphysik, Heidelberg** (Oct 2020 – Dec 2025)
Postdoctoral Researcher
 - Built statistical models on high-dimensional data in Python/C++; ran large-scale parameter scans and quantified uncertainty via rigorous fitting.
- **Vrije Universiteit Brussel, Brussels** (Oct 2018 – Sep 2020)
Postdoctoral Researcher
 - Developed predictive models and a reusable statistical-fitting framework in Python; benchmarked outputs against real data to prioritize hypotheses.
- **Johannes Gutenberg University, Mainz** (Dec 2017 – Sep 2018)
Postdoctoral Researcher
 - Built numerical simulations modeling outcome probabilities across multi-dimensional parameter spaces to identify data-consistent regions.
- **University of Warsaw, Warsaw** (Apr 2016 – Nov 2017)
Research Associate
 - Built and cross-validated computational models combining analytic and numerical methods; verified results via custom simulation and visualization scripts.

Education

- **PhD Physics, University of Warsaw, Warsaw, Poland** (2011 – 2016)
Visiting Scholar at UC Riverside, USA (1 year) and UC Davis, USA (6 months)
- **M.Sc. & M.Phil Physics, Quaid-i-Azam University, Islamabad, Pakistan** (2006 – 2010)
Chancellor Gold Medal

Selected Publications

- Long-Lived-Particle Signals of a Composite Hidden Sector through the Neutrino Portal
A. Ahmed *et al.* – JHEP (2026), [arXiv:2512.09046](#)
Monte Carlo forecasting models (Python/C++) projecting detection sensitivity with statistical confidence bounds under sparse experimental data.
- A Light Dilaton at the LHC
A. Ahmed, A. Mariotti, S. Najjari – JHEP (2020), [arXiv:1912.06645](#)
Statistical fitting algorithms benchmarking predictions against experimental data to map model viability.
- Multi-Component Dark Matter: The Vector and Fermion Case
A. Ahmed, M. Duch, B. Grzadkowski, M. Iglicki – EPJC (2018), [arXiv:1710.01853](#)
Simulated multi-variable coupled dynamics, optimizing convergence across high-dimensional constrained parameter spaces.
- Implications of the Absence of High-Mass Radion Signals
A. Ahmed, B. M. Dillon, B. Grzadkowski, J. F. Gunion, Y. Jiang – PRD (2017), [arXiv:1512.05771](#)
Combined multi-channel data into a joint statistical framework to derive confidence limits – ensemble meta-analysis.

Full list of 30+ peer-reviewed publications: inspirehep.net/authors/1091402

Leadership & Project Management

- **Mentorship & Team Lead:** Co-supervised 10+ PhD and Master's students — from project conception through technical delivery against strict deadlines.
- **Communication & Event Management:** Presented at 50+ international conferences and public science talks; co-organized 10+ conferences/workshops (50+ attendees each).
- **Knowledge Transfer:** Taught advanced physics/mathematics courses at Heidelberg University, VUB Brussels, University of Warsaw, and QAU Islamabad from undergraduate to graduate level.
- **Quality Assurance/Scientific Referee:** Technical referee for JHEP, PRD, and EPJC, verifying mathematical, algorithmic, and data integrity.
- **Cross-Functional Alignment:** Postdoc representative for the *be-h consortium* Belgium, aligning technical roadmaps across 5+ international institutions.